

Assessment of Spalart-Allmaras One-equation BCM Transitional Model using anisotropic metric-based mesh adaptation

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The intent of this work is to investigate whether the Anisotropic Metric-Based mesh adaptation can provide mesh convergence for transitional flows. The model considered is the Spalart-Allmaras BCM Transitional Model. Three test cases of the 1st AIAA CFD Transition Modeling and Prediction Workshop are considered: two-dimensional Zero-Pressure-Gradient Flat Plate; two-dimensional subsonic flow past the NLF-0416 airfoil; three-dimensional subsonic flow past the 6:1 Prolate Spheroid. Attention is dedicated to the mesh-convergence of the transition point, which is automatically detected and meshed with low level of anisotropy. Conversely, high level of anisotropy is observed before and after transition. These requirements permit to significantly limit the number of Degrees of Freedom.

I. Introduction

In the context of CFD calculations, mesh generation for accurate and robust numerical simulations of Reynolds-Averaged Navier-Stokes (RANS) equations is a time-consuming task. This results from the fact that meshes are traditionally considered an input to the simulation pipeline and the common practices consists in manually refining the computational domain according to the *a priori* knowledge of the solution. Unfortunately, this process is not straightforward for complex geometries as well as for flow conditions exhibiting complicated features (e.g., shocks, supersonic shear layers, separation). In this scenario, mesh adaptation strategies based on the interpolation error control are capable of adapting the mesh to the flow solution. The produced adapted meshes exhibit an increased accuracy compared to the standard structured meshes [1] for the same number of degrees of freedom (DoFs). A recent proof of the efficiency of this methodology for complex geometries is the lift prediction for the High-Lift Common Research Model (CRM-HL) provided at the 3rd High-Lift Prediction Workshop [2]. In this application, the same accuracy as "best practice" meshes manually generated is achieved with 75 times fewer vertices for the lift prediction.

Transition to turbulence is one of the most important phenomena in fluid dynamics, which is driven by many different mechanisms and occurs in different applications. Indeed, transition significantly affects flow separation and consequently the performance of any external aerodynamic devices. For this reason, its understanding and the correct modeling is considered a research priority for years to come, as stated in the NASA report "CFD Vision 2030 Study: A Path to Revolutionary Computational Aerosciences" [3]. Direct Numerical Simulations (DNS) or Wall-Resolved Large-Eddy Simulations (WRLES) are still much too expensive to treat turbulent transition occurring in industrial applications. On the other side, the cheaper RANS equations do not have enough turbulence resolution to predict transition to turbulence. For this reason, transition criteria to be coupled with RANS turbulence models have been developed. These criteria are based on: (i) local stability analysis to determine the amplification of instabilities, (ii) local correlations to determine the fraction of time for which the flow is either laminar or turbulent. Both of them are usually coupled with standard turbulence models via an intermittency factor which "turns on/off" the production terms. An example of the first family is the Amplification Factor Transport (AFT) one-equation transition model [4, 5], coupled to the Spalart-Allmaras negative model (SA-neg) [6]. Conversely, the more famous Langtry-Menter $\gamma - Re_\theta$ [7] belongs to the second approach, traditionally coupled to Menter's 2003 Shear-Stress Transport (SST) turbulence model [8] and more recently also to the SA-neg model [9, 10]. Particularly, the Langtry-Menter $\gamma - Re_\theta$ model coupled either with SA or Menter's SST turbulence model, was not able to exhibit mesh convergence on two-dimensional NASA NLF-0416 (Natural Laminar Flow) airfoil on "best practice" meshes up to 32.8 million points [11]. Therefore, the interest in using mesh adaptation for transitional flows is clear and could significantly reduce the number of DoFs as well as limit the human intervention in the mesh-generating process.

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Preliminary attempts in mesh-adapted calculations using either the AFT two-equation model or the Langtry-Menter $\gamma - Re_\theta$ 4-equation model did not produce any satisfactory results. In the former, no mesh-convergence was achieved due to poor numerical behavior of the model [12]. In the latter, no mesh convergence was achieved until an extremely fine resolution i.e., 4 000 000 vertices [13]. In this work, we propose an alternative strategy for transitional flows: the Spalart-Allmaras One-equation BCM transitional model. We demonstrate that this model is robust and provides mesh-convergence solution with a reasonable amount of DoFs.

The paper is organized as follows. A concise presentation of the RANS equations using the baseline Spalart-Allmaras (SA) turbulence model and the transitional SA-BCM turbulence model is introduced in Section II. Section III presents the discretization adopted as well as the main pillars of the metric-based mesh adaptation. Section IV is dedicated to two-dimensional and three-dimensional adapted calculations. Concluding remarks are given in Section V.

II. Modeling equations

A. Navier-Stokes equations

The compressible Navier-Stokes equations for mass, momentum and energy conservation read:

$$\left\{ \begin{array}{l} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \\ \frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = \nabla \cdot \mathcal{T}, \\ \frac{\partial (\rho E)}{\partial t} + \nabla \cdot ((\rho E + p) \mathbf{u}) = \nabla \cdot (\mathcal{T} \cdot \mathbf{u} - Q), \end{array} \right. \quad (1)$$

where ρ denotes the density, $\mathbf{u}$ the velocity, p the static pressure and E the total specific energy. The latter is defined as the sum of the flow specific internal energy e and the specific kinetic one as

$$E = e + \frac{\|\mathbf{u}\|^2}{2}.$$

The fluid is assumed to be Newtonian and the viscous stress tensor $\mathcal{T}$ is defined by

$$\mathcal{T} = \mu \left((\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} \nabla \cdot \mathbf{u} \mathbb{I} \right) = \mu \boldsymbol{\tau}, \quad (2)$$

where μ denotes molecular dynamic viscosity and represents the resistance for unit area of a fluid to a unit deformation by shear stress. For a perfect gas, μ strongly depends on temperature but only weakly on pressure. For this reason, fluid is imposed to obey Sutherland's law, i.e.

$$\mu = \mu_\infty \left(\frac{T}{T_\infty} \right)^{\frac{3}{2}} \left(\frac{T_\infty + \text{Su}}{T + \text{Su}} \right)$$

where $\text{Su} = 110.4 \text{ K}$ is the Sutherland temperature for dry air and the index ∞ denotes reference quantities.

The heat transfer due to temperature gradient is provided by Q and reads a form of Fourier's law of heat conduction, i.e.,

$$Q = -\lambda \nabla T$$

with λ standing for the thermal conductivity coefficient and T for the absolute static temperature. In the case of gases, λ depends on temperature in a similar way as μ . For this reason, the following relationship is accepted

$$\lambda = c_p \frac{\mu}{\text{Pr}} \quad \text{with} \quad \text{Pr} = 0.72 \quad \text{for (dry) air.}$$

Assuming the fluid to be a calorically perfect gas, the closing relations are

$$p = \rho R T, \quad e = c_v T \quad (3)$$

where $R = \mathcal{R}/M$ is the perfect gas constant divided by the molar mass of the fluid, commonly set at $287.1 \text{ J kg}^{-1} \text{ K}^{-1}$ for air and c_v is the specific heat capacity at constant volume. The latter is linked to specific heat capacity at constant pressure c_p by the specific heat ratio $\gamma = c_p/c_v$, set at 1.4 for diatomic gas.

B. RANS equations

In the case of the Reynolds Average Navier-Stokes (RANS) equations, the flow variables are decomposed into a time-average and a fluctuating part, and only the governing equations for the former are solved. Particularly, density and pressure are only time averaged, while the other variables undergo an additional mass-weighted averaging. The consequent equations' system is equivalent to that in Eq. (1), except for the terms $\mathcal{T}_t$ and Q_t , called Reynolds stress tensor and turbulent heat flux respectively, which appear added to $\mathcal{T}$ and Q in the momentum and energy equations. These extra terms are defined via the *Boussinesq* hypothesis as

$$\mathcal{T}_t = \mu_t \tau \quad \text{and} \quad Q_t = -\lambda_t \nabla T, \quad (4)$$

where μ_t is called turbulent dynamic viscosity. The turbulent conductivity λ_t is expressed from the Prandtl turbulent number:

$$\lambda_t = \frac{\mu_t c_p}{\text{Pr}_t} \quad \text{with} \quad \text{Pr}_t = 0.9 \quad \text{for (dry) air}.$$

In order to close the system, μ_t is calculated by means of a turbulence model.

Turbulence modeling: Negative Spalart-Allmaras One-Equation Model without ft2 Term (SA-neg-noft2) The turbulence formulation considered here is the Negative Spalart-Allmaras turbulence model with neglecting trip terms (SA-neg-noft2) [6]. It consists in a convection-diffusion equation for the eddy viscosity variable $\tilde{\nu}$ where the source term is obtained as a balance between a production term and a destruction term associated with the same quantity, as

$$\underbrace{\frac{\partial \rho \tilde{\nu}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \tilde{\nu})}_{\text{convection}} - \underbrace{\frac{\rho}{\sigma} \nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu})}_{\text{diffusion}} = \underbrace{\frac{c_{b2} \rho}{\sigma} \|\nabla \tilde{\nu}\|^2}_{\text{diffusion}} + \underbrace{c_{b1} \tilde{S} \rho \tilde{\nu}}_{\text{production}} - \underbrace{c_{w1} f_w \rho \left(\frac{\tilde{\nu}}{d}\right)^2}_{\text{destruction}}. \quad (5)$$

The turbulent viscosity is computed from

$$\mu_t = \rho \tilde{\nu} f_{v1},$$

where

$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \quad \text{and} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{with} \quad \nu = \frac{\mu}{\rho}.$$

Additional definitions are given by the following equations

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{and} \quad \tilde{S} = \Omega + \bar{S}$$

where d is the distance to nearest wall, $\Omega = \|\nabla \times \mathbf{u}\|$ and $\bar{S} = \tilde{\nu}/(\kappa^2 d^2) f_{v2}$. In order to avoid numerical problems when $\tilde{S} \leq 0$, the following correction proposed by Allmaras and Johnson [14], is adopted:

$$\tilde{S} = \Omega + \frac{\Omega (c_2^2 \Omega + c_3 \bar{S})}{(c_3 - 2c_2) \Omega - \bar{S}} \quad \text{when} \quad \bar{S} < -c_2 \Omega,$$

with $c_2 = 0.7$ and $c_3 = 0.9$.

The set of closure constants for the model is given by

$$\begin{aligned} \sigma &= 2/3, & c_{b1} &= 0.1355, & c_{b2} &= 0.622, & \kappa &= 0.41, & c_{t3} &= 1.2, \\ c_{w1} &= c_{b1}/\kappa^2 + (1 + c_{b2})/\sigma, & c_{w2} &= 0.3, & c_{w3} &= 2, & c_{v1} &= 7.1, & c_{t4} &= 0.5. \end{aligned}$$

Finally, the function f_w is computed as:

$$f_w = g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6} \quad \text{with} \quad g = r + c_{w2} (r^6 - r) \quad \text{and} \quad r = \min \left(\frac{\tilde{\nu}}{\tilde{S}_\kappa^2 d^2}, 10 \right).$$

When $\tilde{\nu}$ becomes negative, the SA model is referred as SA-neg [14] and Eq. (5) is replaced by the following

$$\frac{\partial \rho \tilde{\nu}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \tilde{\nu}) - \frac{\rho}{\sigma} \nabla \cdot ((\nu + f_n \tilde{\nu}) \nabla \tilde{\nu}) = \frac{c_{b2} \rho}{\sigma} \|\nabla \tilde{\nu}\|^2 + c_{b1} (1 - c_{t3}) \Omega \rho \tilde{\nu} - c_{w1} \rho \left(\frac{\tilde{\nu}}{d} \right)^2 \quad (6)$$

with

$$f_n = \frac{16 + \chi^3}{16 - \chi^3}.$$

Transitional model: Spalart-Allmaras One-Equation BCM Transitional Model (SA-neg-noft2-BCM) [15] The transitional model considered in this work relies on a modification of the production term of the SA-neg-noft2, which is multiplied with a γ_{BC} intermittency function. This function damps any turbulence production until some transition criteria is achieved. The resulting turbulence model is

$$\begin{aligned} \frac{\partial \rho \tilde{\nu}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \tilde{\nu}) - \frac{\rho}{\sigma} \nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) &= \frac{c_{b2} \rho}{\sigma} \|\nabla \tilde{\nu}\|^2 + c_{b1} \gamma_{BC} \tilde{S} \rho \tilde{\nu} - c_{w1} f_w \rho \left(\frac{\tilde{\nu}}{d} \right)^2 & \tilde{\nu} > 0 \\ \frac{\partial \rho \tilde{\nu}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \tilde{\nu}) - \frac{\rho}{\sigma} \nabla \cdot ((\nu + f_n \tilde{\nu}) \nabla \tilde{\nu}) &= \frac{c_{b2} \rho}{\sigma} \|\nabla \tilde{\nu}\|^2 + c_{b1} \gamma_{BC} \rho \|\nabla \times \mathbf{u}\| + c_{w1} \rho \left(\frac{\tilde{\nu}}{d} \right)^2 & \tilde{\nu} < 0 \end{aligned}$$

with

$$\gamma_{BC} = 1 - \exp \left(-\sqrt{T_1} - \sqrt{T_2} \right).$$

T_1 and T_2 are triggering functions defined as

$$T_1 = \frac{\max (Re_\theta - Re_{\theta,c}, 0)}{\chi_1 Re_{\theta,c}}, \quad T_2 = \max \left(\frac{\mu_T}{\chi_2 \mu}, 0 \right),$$

with $\chi_1 = 0.002$ and $\chi_2 = 0.02$.

The transition onset is based on the following experimental correlation

$$Re_{\theta,c} = 803.73 (Tu_\infty + 0.6067)^{-1.027}, \quad (7)$$

where Tu_∞ is the free-stream turbulence intensity and Re_θ is based on

$$Re_\theta = \frac{Re_\nu}{2.193}, \quad Re_\nu = \frac{\rho d^2 \|\nabla \times \mathbf{u}\|}{\mu}.$$

The free-stream boundary condition is

$$\frac{\tilde{\nu}_\infty}{\nu_\infty} = 0.02.$$

Pay attention to the fact that turbulence intensity must be specified in percent i.e., if it is about 0.18% you need to set $Tu_\infty = 0.18$.

III. Numerical method

The flow solver considered in this work is WOLF [11, 16–18], a vertex-centered Mixed Finite Volume - Finite Element Reynolds Averaged Navier-Stokes (RANS) solver for unstructured meshes. This means that the convective fluxes are solved by the Finite Volume method on a dual mesh composed of median cells, while the viscous fluxes are evaluated using the Finite Element method. Specifically, an HLLC approximate Riemann solver [19] is used in concert with a low dissipation second order MUSCL scheme using a combination of centered (edge gradient) and upwind gradients (Finite Element gradient), called V4 scheme [20, 21]. Similarly, the Finite Element Method guarantees a second order accuracy for the viscous terms when simplicial elements (triangles and tetrahedra) are adopted.

The time integration considered here is the backward Euler time-integration scheme for implicit temporal discretization. Specifically, at each time step the RANS equations are approximately solved using a Symmetric Gauss–Seidel (SGS) method with a local time stepping. An implicit loosely coupled algorithm is used to integrate the mean-flow equations and Spalart–Allmaras turbulence equation separately [22, 23].

A. Mesh adaptation

In the mesh adaptation process, the local size and the anisotropy of the mesh are prescribed using a metric field [11, 24]. In this context, a tetrahedron is said *unit* according to a metric if all its edges have unit length in this metric. Similarly, a mesh which is unit for a given metric field, is composed of unit elements in this metric. This condition establishes a duality between meshes and metrics: a Riemannian metric field can be seen as the continuous counterpart of a mesh. The reader can refer to Loseille and Alauzet [25, 26] for more details. In this way, the mesh generation process is recast in generating a uniform unit mesh in the prescribed metric space.

Denoting by $\mathcal{M}_{\text{Opt}}$ the optimal metric, $C(\mathcal{M}_{\text{Opt}})$ its complexity, *i.e.* the continuous counterpart to mesh size, $\mathcal{N}$ a target complexity, and $\mathcal{E}$ the error model for the considered application, the problem to solve is

$$\mathcal{M}_{\text{Opt}} = \underset{C(\mathcal{M})=\mathcal{N}}{\operatorname{argmin}} \mathcal{E}(\mathcal{M}).$$

The resulting non-linear process is depicted in Algorithm 1. The optimal metric field $\mathcal{M}_{\text{Opt}}$ used to prescribe the new adapted mesh $\mathcal{H}$ is automatically deduced from the actual solution or from the actual solution and adjoint state with different error estimates [11]. In practice, an additional step is required between the metric computation step and the mesh generation step, called metric gradation. Indeed, metrics computed from numerical solutions are likely to show irregularities and need to be smoothed, through a gradation process, to improve the quality of the adapted mesh [27, 28].

Algorithm 1: General mesh adaptation loop with mesh-convergence analysis

Input: Initial mesh $\mathcal{H}_0^0$, solution W_0^0 , and complexity C^0

while ($C^j \leq C^{\max}$) **do**

Inner loop to converge the mesh adaptation at fixed complexity:

1) **while** ($i \leq n_{\text{adap}}$) **do**

1) Compute the optimal metric for the considered error estimate and complexity $\rightarrow \mathcal{M}_{i-1}^{j,0}$;

2) Apply a gradation correction to smooth the metric field $\rightarrow \mathcal{M}_{i-1}^j$;

3) Generate a new adapted mesh $\rightarrow \mathcal{H}_i^j$;

4) Interpolate the solution on the new mesh $\rightarrow (W^0)_i^j$;

5) Compute the solution $\rightarrow W_i^j$;

6) **if** (*Convergence check*) **then** the loop has converged

$i = n_{\text{adap}} + 1$;

else the loop has not converged

$i = i + 1$;

end

end

2) $\mathcal{H}_0^{j+1} = \mathcal{H}_{n_{\text{adap}}}^j$; $W_0^{j+1} = W_{n_{\text{adap}}}^j$; $C^{j+1} = \alpha \cdot C^j$;

end

Error estimate. The flow solver WOLF can provide metrics for either *multi-scale* or *goal-oriented* error estimates. The *multi-scale* error estimate, also known as Feature-Based (FB), aims at minimizing the interpolation error in the L^p norm of a given sensor for a given number of degrees of freedom. It is easy to implement because it only requires the Hessian matrix of the sensor. The *goal-oriented* error estimate (GO) minimizes the error of a given engineering output functional for a given number of degrees of freedom. It is more complex because it requires a robust adjoint solver and a proper differentiation of the output functional [1].

The objective of our study is the assessment of an anisotropic mesh-converged solution for the SA-neg-noft2-BCM transitional model. All the adaptive simulations shown in this work are performed using the mesh-adaptive solution platform composed of WOLF, the anisotropic local remeshing software FEFLO.A [29-31] and the field interpolator INTERPOL [32, 33]. Each step of the adaptation process is detailed in [2]. All the solutions are visualized with Vizir4 [34].

IV. Numerical results

A. 2D Zero-Pressure-Gradient Flat Plate

The first test case considered is the subsonic flow over a 2D flat plate with zero pressure gradient. The flow conditions are: Mach number $M = 0.1443$ and Reynolds number $Re = 3.34 \times 10^6$, with the latter based on the unit flat plate length. The Turbulence Intensity specified at free-stream is $Tu_\infty = 0.18\%$. This case is the one considered by Schubauer and Klebanoff [35] and was already used to calibrate the BCM-model [36].

The initial mesh considered is shown in Fig. 1. This is based on the coarsest structured grid available on the NASA Langley Research Center website* and obtained by splitting all the quadrilaterals along the same diagonal. The inflow boundary is placed at $x = 0.3$ from the flat plate leading edge and the flat plate length is about $L = 2.5$. Boundaries are highlighted with different colors. The total pressure for undisturbed isentropic flow is imposed at the inflow (yellow reference), while subsonic flow with unit static pressure ratio is imposed at the outflow (magenta reference). On the flat plate surface (red reference), an adiabatic no-slip boundary condition is imposed. A symmetry and farfield Riemann boundary condition is imposed on the bottom (green reference) and top (gray reference) boundaries, respectively.

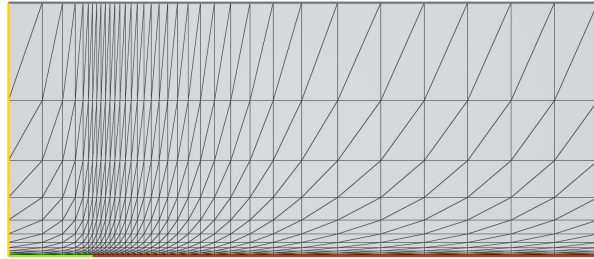


Fig. 1 2D Zero-Pressure-Gradient Flat Plate verification test: Initial coarse mesh only composed of 875 vertices, 1632 triangles and 116 edges on the boundaries.

We consider a multi-scale error (FB) estimate [2, 18, 37] based on the L^4 norm of the Mach number field. The flow solver calculation is stopped when the residuals fall below 1×10^{-8} . The refinement level of each mesh is based on a number called complexity C (see Section III), which is proportional to the number of vertices. A maximum number of 20 mesh adaptations at each complexity is imposed. The exit criterion for each complexity is based on the percentage difference of the drag coefficients between two subsequent adaptations and set to $\epsilon_{CD} = 0.001$.

* NASA Langley Research Center, 2D Zero Pressure Gradient Flat Plate Verification Case, Turbulence Modeling Resource, <https://turbmodels.larc.nasa.gov/flatplate.html>

Assessment of an anisotropic metric-based mesh-converged solution with FB error estimate First, a fully turbulent RANS adaptive process using SA-neg-noft2 model is performed.

A mesh-convergence study is obtained on the following eight complexities:

$$C = \{ 500, \quad 1000, \quad 2000, \quad 4000, \quad 8000, \quad 16\,000, \quad 32\,000, \quad 64\,000 \}.$$

For these complexities, the final adapted meshes size results between 600 and 75 000 vertices. Second, the SANeg-noft2 fully turbulent solution obtained on a 2000-vertices adapted mesh is used to initialize the SANeg-noft2-BCM transitional calculation. The evolution of drag coefficient C_D against the number of vertices N is depicted in Fig. 2, on the left. It is interesting to note that the fully turbulent solution is already sufficiently converged at about 10 000 vertices. Conversely, the transitional solution seems to require more points, i.e. approximately 20 000 vertices. In the next paragraph, we will see that this aspect is likely related to the transition point capturing. In the same picture, on the right, the skin friction coefficient C_f obtained on the final 20K-vertices adapted meshes is shown for both adaptations. The reader can clearly see, for the transitional simulation, the well-known laminar behavior up to $Re_x \approx 3 \times 10^6$ at which, in accordance with literature, transition occurs and the profile approaches the turbulent one.

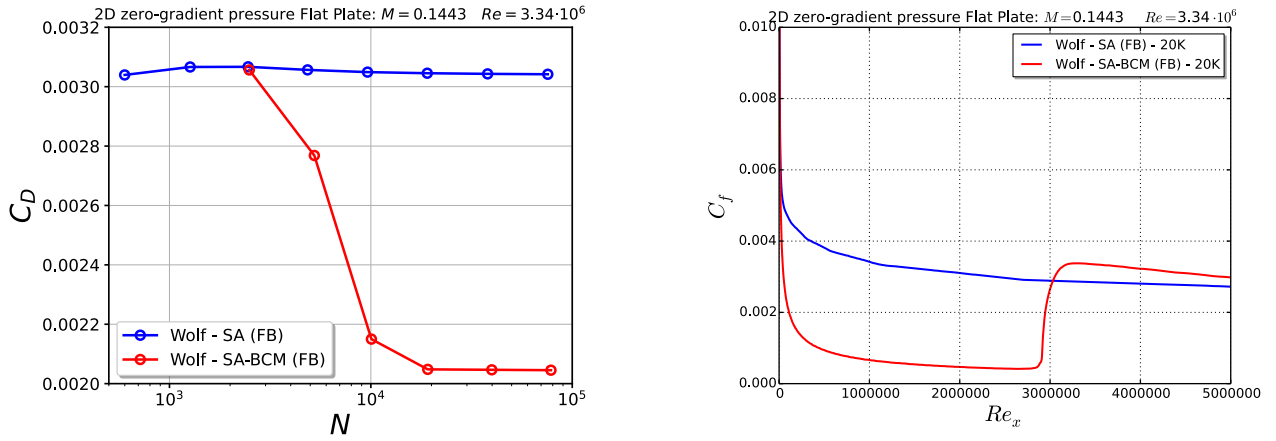


Fig. 2 2D Zero-Pressure-Gradient Flat Plate FB anisotropic metric-based mesh adaptation. (left) Evolution of the drag C_D coefficient during the mesh-convergence process. N is the number of vertices of the adapted meshes. (right) Skin friction C_f coefficient on a 20 000 vertices-adapted meshes. Comparison between fully turbulent (SANeg-noft2) and transitional (SANeg-noft2-BCM) calculation.

Transition point mesh-convergence Our main objective here is to assess the mesh-convergence for the SA-BCM model, specifically in terms of the position of the transition point. Fig. 3, shows that during the adaptation process, the transition point does not oscillate but converge to a specific position. This means that this model is robust and guarantees mesh-convergence solutions. Particularly, the reader can note that the transition points does not change anymore since 20 000 vertices.

The resulting adapted meshes are shown in Fig. 4. On the left, the final adapted mesh and solution for the SA model. On the right, the final adapted mesh and solution for the SA-BCM model. Both meshes are stretched along the wall-normal direction by a factor 20, to highlight the leading edge and the transitional region. We can see that the adaptation process captures automatically all discontinuities in the flow. Specifically, the leading edge is a discontinuity in terms of boundary condition. The transition point is a discontinuity in the solution, when passing from laminar to turbulent flow.

A close view of the transition point is shown in Fig. 5. Both meshes are stretched along the wall-normal direction by a factor 5, to highlight the refinement close to the transition point. We can see in detail as the adaptation process provides the proper number of points according to the complexity of the flow. The laminar boundary layer requires much less points w.r.t. the turbulent one, to be correctly captured. Then, we can see the transition to turbulence already in the mesh: the upstream region for the SA-BCM calculation is laminar, velocity gradients are lower and boundary layer is thin. Downstream, velocity gradients are higher and boundary layer is thick. Such physical considerations are well

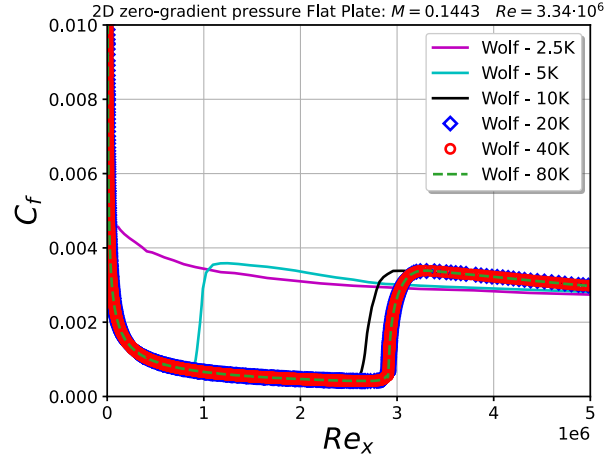


Fig. 3 2D Zero-Pressure-Gradient Flat Plate. FB anisotropic metric-based mesh adaptation using the SANeg-noft2-BCM model. Evolution of the skin friction C_f coefficient during the adaptation process. The mesh-convergence is achieved at 20 000 vertices.

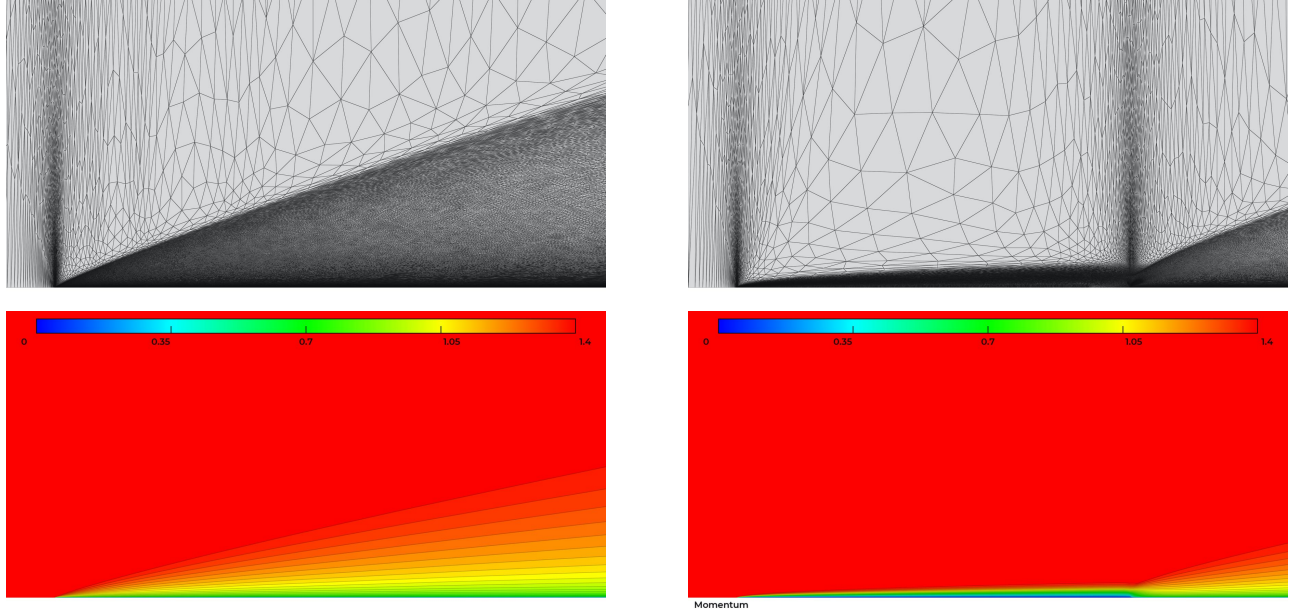


Fig. 4 2D Zero-Pressure-Gradient Flat Plate. FB anisotropic metric-based mesh adaptation using the SANeg-noft2-BCM model. (left) Adapted mesh and streamwise velocity component field for turbulent solution. (right) Adapted mesh for and streamwise velocity component field for transitional solution. Both meshes are stretched along the wall-normal direction by a factor 20, to highlight the leading edge and the transitional region.

highlighted by the refinement region. In the transition region, adaptation becomes more isotropic to capture transition. Conversely, it is still highly anisotropic before and after the transition region.

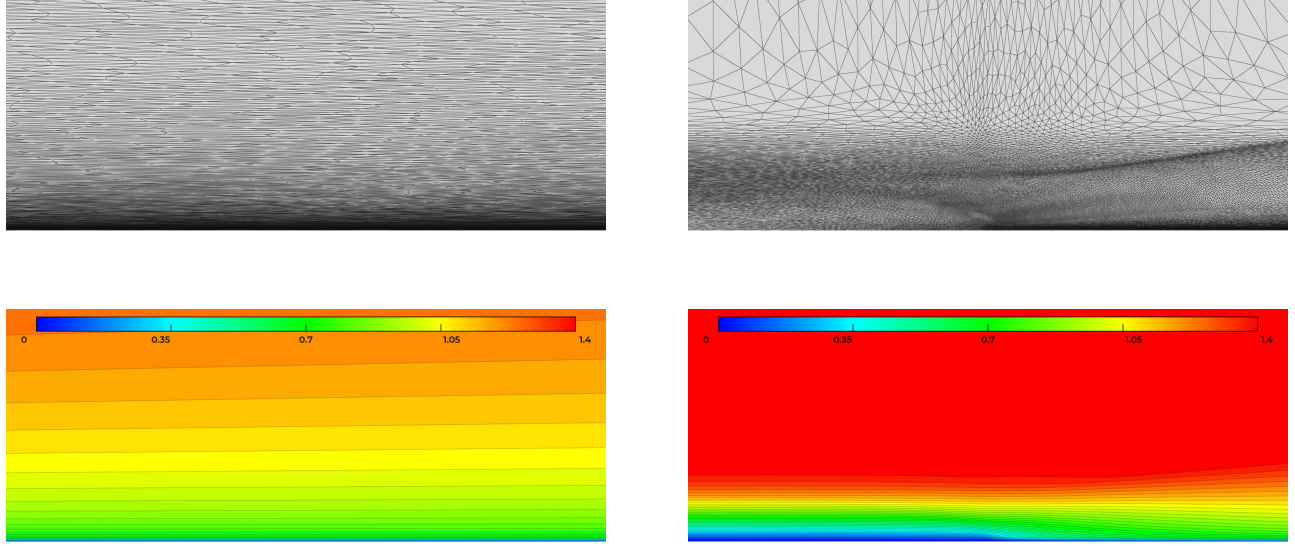


Fig. 5 2D Zero-Pressure-Gradient Flat Plate FB anisotropic metric-based mesh adaptation. (left) Adapted mesh for turbulent solution. (right) Adapted mesh for transitional solution. Both meshes are stretched along the wall-normal direction by a factor 5, to the transitional region.

B. 2D subsonic flow past the NLF4016 airfoil

The case considered is the flow around the NLF4016 airfoil at Mach number $M = 0.1$ and Reynolds number $Re = 4 \times 10^6$ based on the airfoil chord length c . Several experimental data are available for Re ranging from 1×10^6 to 9×10^6 , and M from 0.1 to 0.4 [38]. In these experiments a specific roughness was opportunely sized for each Reynolds number to fix the transition point at $0.075c$ on both surfaces.

It should be stressed that an-induced transition by roughness is totally different by a by-pass one induced by a fixed free-stream turbulence. Therefore, there is no guarantee that the transition point is the same as that measured in the experiments. This aspect is crucial, because here we don't want to do a validation of the numerical simulation with experiments, but rather analyzing the robustness of the model for mesh-adaptation applications.

Assessment of an anisotropic metric-based mesh-converged solution with GO error estimate The NLF4016 airfoil is run with the goal-oriented error estimate using the drag as the targeted functional. The SA-BCM model, with fixed free-stream turbulence intensity Tu_∞ is used to calculate the adapted RANS solution. The free-stream temperature is set at $T_\infty = 300$ K and the free-stream Turbulence Intensity at $Tu_\infty = 0.15$ %, according to the 1st AIAA Transition Modeling Workshop [4].

Two different mesh-convergence study are run, with a maximum number of 15 mesh adaptations at each complexity. The exit criterion for each complexity is based on the percentage difference of drag coefficient between two subsequent adaptations and set at $\epsilon = 5 \times 10^{-5}$. The study which is fully turbulent and based on baseline SA model is performed on the six complexities:

$$C^{SA} = \{ 1000, \quad 2000, \quad 4000, \quad 8000, \quad 16\,000 \quad 32\,000 \}.$$

For these complexities, the final adapted meshes size results between 1200 and 40 000 vertices.

[†] AIAA Transition Modeling Workshop, Natural-Laminar-Flow Common Research Model (CRM-NLF), AIAA Transition Modeling Workshop-I, https://transitionmodeling.larc.nasa.gov/workshop_i/?doing_wp_cron=1716393421.7890670299530029296875

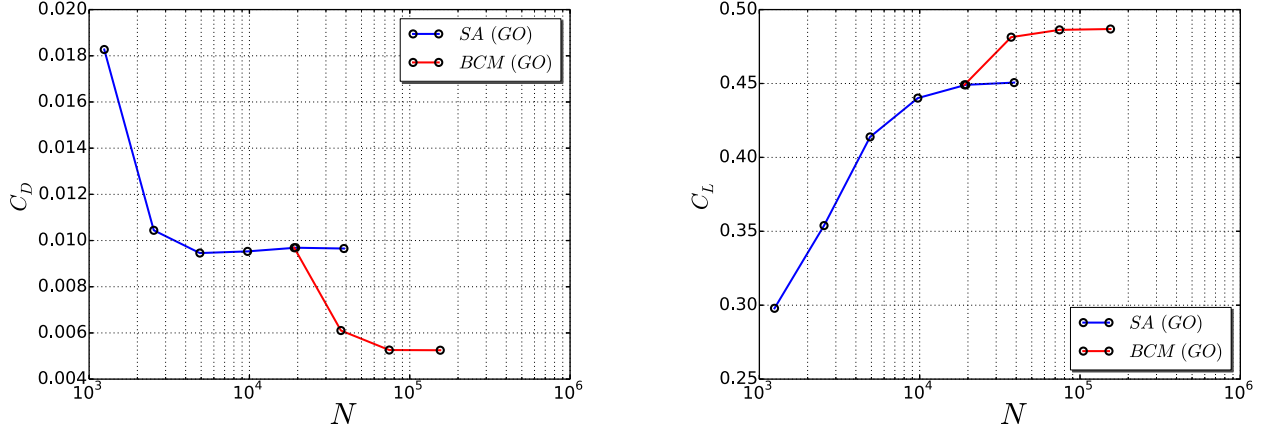


Fig. 6 2D flow past the NLF4016 airfoil at $\text{AoA} = 0^\circ$, $Tu_\infty = 0.15$. GO anisotropic metric-based mesh adaptation. Evolution of the Drag C_D coefficient during the adaptation process. Comparison between fully turbulent calculation (blue curve) and transitional calculation (red curve).

The transitional simulation using the BCM model is run on the four complexities:

$$C^{\text{BCM}} = \{ 16\,000, \quad 32\,000, \quad 64\,000, \quad 128\,000 \}.$$

For these complexities, the final adapted meshes size results between 20 000 and 150 000 vertices.

As previously done for the 2D flat plate calculation, the adaptive process for the SA-BCM model is initialized with an adapted mesh and solution obtained with the SA model. For this study, the one obtained with 20 000 vertices (i.e., $C^{\text{SA}} = 16\,000$) is used.

The convergence for the drag coefficient C_D and lift coefficient C_L against the number of vertices N , is plotted in Fig. 6. We can see that using anisotropic mesh adaptation only 20 000 vertices are needed to obtain the mesh-convergence with SA model. Conversely, 75 000 vertices are needed for the BCM model.

To understand why more points are needed for the transitional simulations, consider Fig. 7. Mesh and x-component of the velocity field obtained with SA model and BCM model are depicted on the left and right, respectively. Both meshes are stretched along the y-direction by a factor 5, to stress the transitional region. We can see that a fully turbulent calculation leads to high anisotropy everywhere on the wall surface. In fact, high gradients needs low size to be properly captured. Conversely, laminar region does not need so tiny sizes. Nevertheless, when transition occurs the domain is meshed quasi-isotropically and becomes again highly stretched downstream the transition point. Moreover, an additional refinement layer seems to be present in the adapted mesh obtained with the BCM model. The origin can be traced by looking at Fig. 8. We can see that the apparent additional refinement is related to the delayed development of the turbulent viscosity. Transitional simulations, being characterized by a laminar regime, do not need any turbulence modeling before transition. Thus, no turbulent viscosity is produced. Downstream transition, turbulence occurs and turbulence viscosity is produced. The edge of the boundary layer is detected by the error estimate and refined. This occurs because here the gradient of the pseudo-turbulent viscosity $\tilde{\nu}$ is discontinuous, as pointed out in Ref [39].

Transition point mesh-convergence Fig. 9 shows the convergence of the transition point during the adaptation process. As for the 2D flat plate, the transition point converges to a specific location and it does not change anymore since the 75 000 vertices-adapted mesh. This mesh-convergence is achieved for all the AoAs investigated.

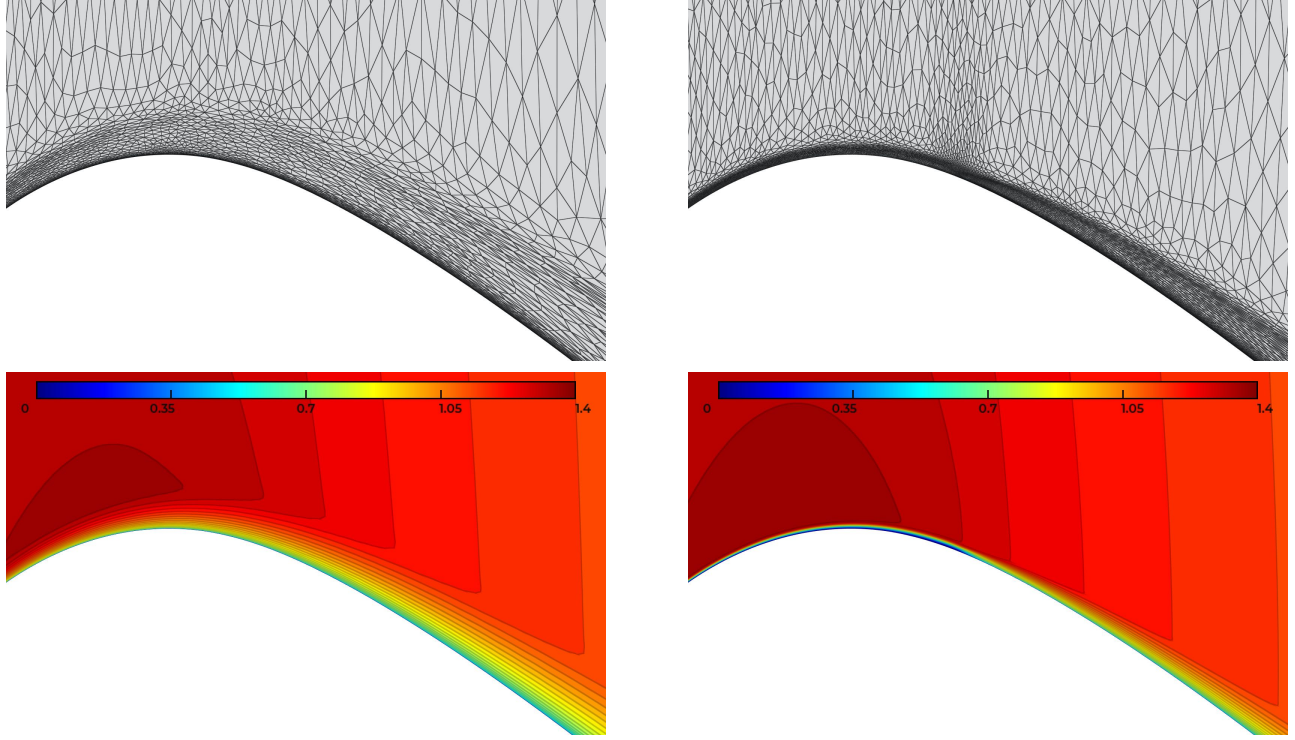


Fig. 7 2D flow past the NLF4016 airfoil at $\text{AoA} = 0^\circ$, $Tu_\infty = 0.15$. GO anisotropic metric-based mesh adaptation. Adapted mesh obtained with the SA-neg-noft2 model and composed of 40 000 vertices (top left). Adapted mesh obtained with the SA-neg-noft2-BCM model and composed of 75 000 vertices (top right). Streamwise velocity component field (bottom). Both meshes are stretched by a factor 5, along the y- direction.

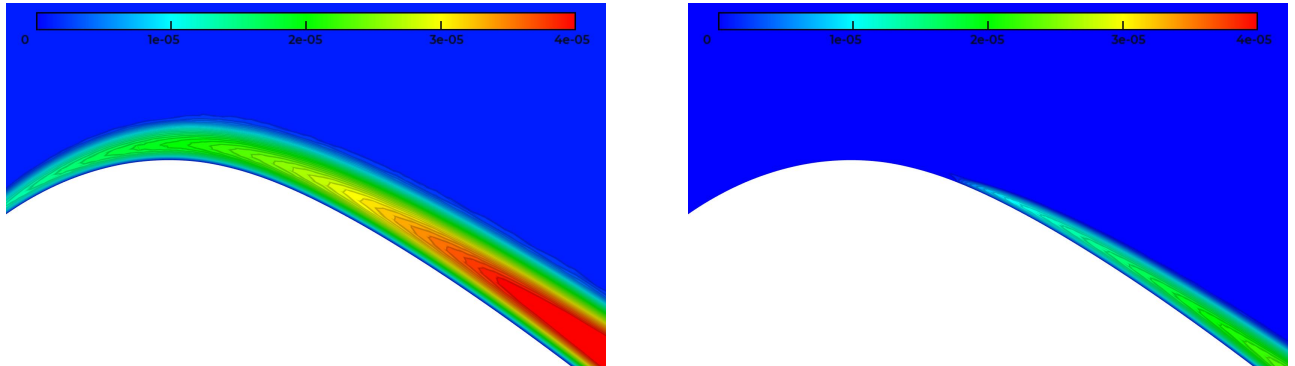


Fig. 8 2D flow past the NLF4016 airfoil at $\text{AoA} = 0^\circ$, $Tu_\infty = 0.15$. GO anisotropic metric-based mesh adaptation. Pseudo-turbulent viscosity $\bar{\nu}$ field obtained for a fully turbulent calculation on 40 000 vertices adapted mesh (left) and transitional calculation on 75 000 vertices adapted mesh (right). Both meshes are stretched by a factor 5, along the y- direction.

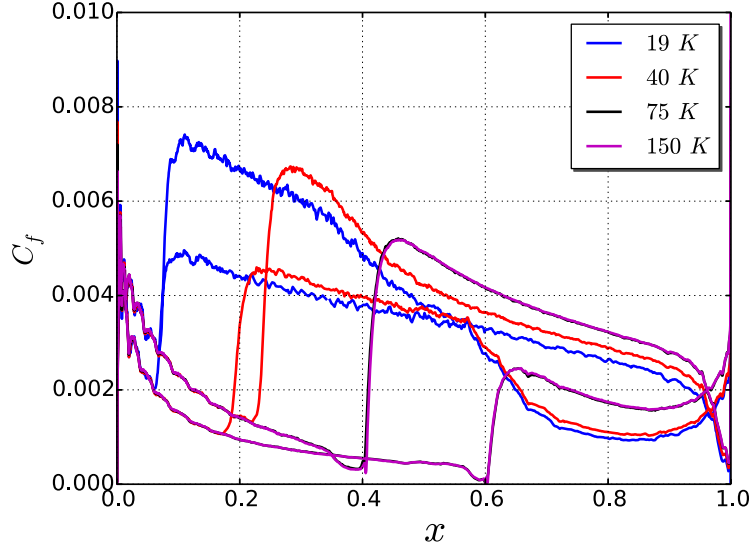


Fig. 9 2D flow past the NLF4016 airfoil at $AoA = 0^\circ$, $Tu_\infty = 0.15$. GO anisotropic metric-based mesh adaptation. Evolution of the skin friction coefficient C_f for different mesh sizes, during the adaptation process. The solution is mesh-converged at the 75 000 vertices adapted mesh.

Transition at higher Angles of Attack The skin friction coefficient C_f obtained with adaptive calculations at different Angles of Attack (AoAs) is shown in Fig. 10. In order to speed-up convergence, each adaptive simulation is initialized with the final couple mesh-solution obtained for the previous AoA, using the α -continuation method [40] with one-degree increment. The process is confirmed to be robust. It is interesting to note that for $Tu = 0.15$, the upper and lower surface still have different point of transition for different AoA. Specifically, the reader can observe that the upper profile transition earlier and earlier, while the lower profile later and later.

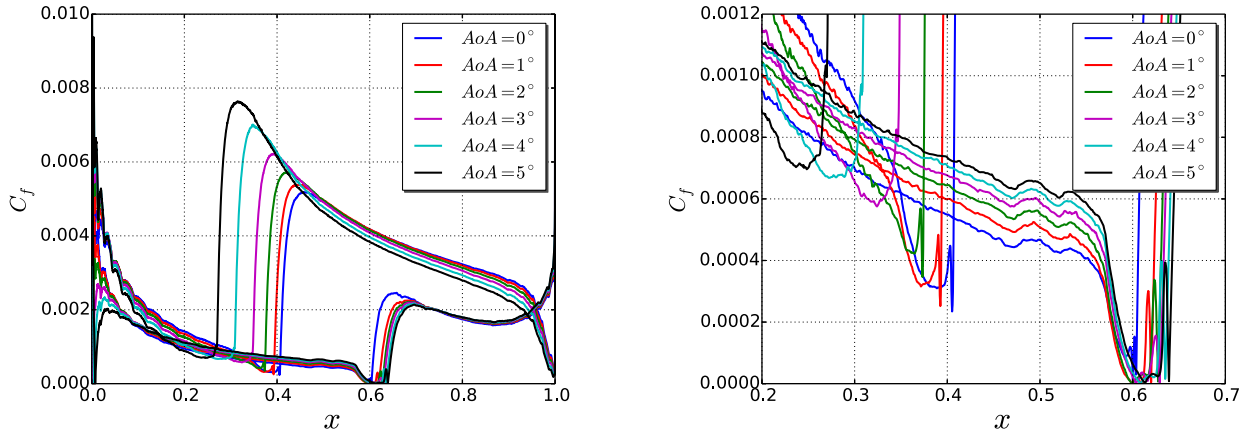


Fig. 10 2D flow past the NLF4016 airfoil at $AoA = 0^\circ$. GO anisotropic metric-based mesh adaptation. Evolution of the Skin Friction C_f for different AoAs. The solution considered is obtained with mesh-converged at the 150 000 vertices adapted mesh.

Transition for different free-stream turbulence intensity An important aspect is the sensitivity of the model to different free-stream turbulence intensities Tu_∞ . The left picture in Fig. 11 shows the behavior of the model for different Tu_∞ . We can see that the transition occurs earlier and earlier with increasing turbulence intensity. Specifically, the lower surface seems to be more sensitive than the upper one. In fact, from a value of $Tu_\infty = 0.005$ to $Tu_\infty = 0.25$ the difference in the locations of the transition points on both surfaces is fewer and fewer. Another important remark is that below $Tu_\infty = 0.01$, transition points do not change anymore. This result is quite evident when looking at Eq. (7). The trend of the critical momentum thickness Reynolds number $Re_{\theta,c}$ is plotted against Tu_∞ in the same figure on the right. The reader can easily see that a limit value of about 1342.75 is obtained for infinitesimal turbulence intensity.

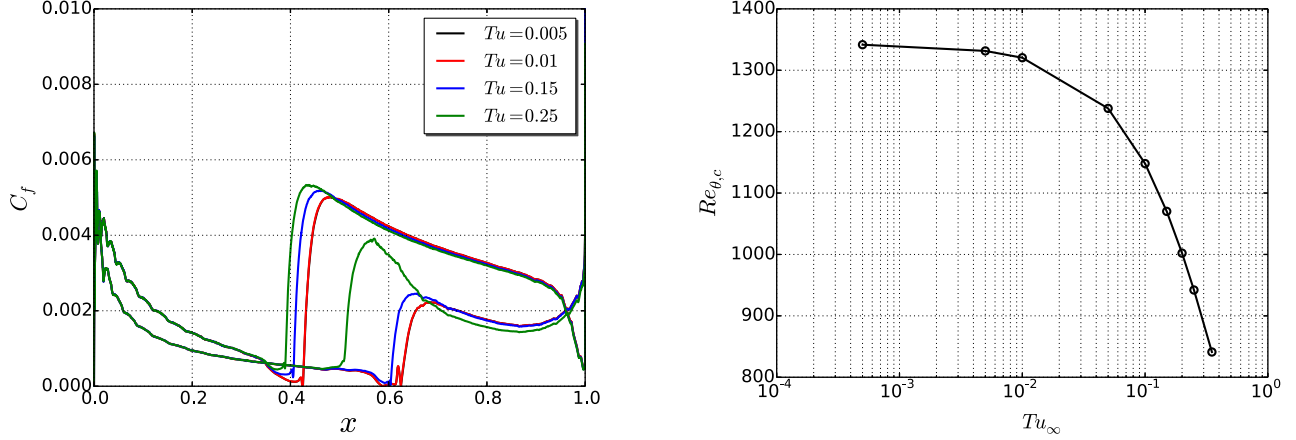


Fig. 11 2D flow past the NLF4016 airfoil at $AoA = 0^\circ$. GO anisotropic metric-based mesh adaptation. Evolution of the Skin Friction C_f coefficient for different free-stream turbulent intensity Tu_∞ on 150 000 vertices adapted mesh(left). Behavior of $Re_{\theta,c}$ for different free-stream turbulent intensity Tu_∞ (right).

C. 3D subsonic flow past the 6:1 Prolate Spheroid

The three-dimensional case considered is the subsonic flow past the 6:1 Prolate Spheroid at Mach number $M = 0.13$ and Reynolds number $Re = 6.5 \times 10^6$ based on the major axis of the spheroid b . This case was proposed as one of the three-dimensional simulations demanded for the 1st AIAA Transition Modeling Workshop[†]. In this paper, the SA-BCM model with fixed free-stream turbulence intensity Tu_∞ is used to calculate the adapted RANS solution. The free-stream temperature is set at $T_\infty = 300$ K, the angle of attack $AoA = 5^\circ$ and the free-stream Turbulence Intensity $Tu_\infty = 0.15\%$ [†]. The initial mesh considered is shown in Fig. 12.

As previously done, two different mesh-convergence study are performed, with a maximum number of 15 mesh adaptations at each complexity. One using the baseline SA-neg model and the other using the BCM-SA-neg model. The error estimate considered is the goal-oriented estimate based on the drag coefficient. Convergence for each mesh is considered achieved when non-linear residuals fall below $R_{lg} = 1 \times 10^{-8}$. At this point, the flow solver calculation is stopped and a metric field is computed.

The exit criteria for each complexity are based on the percentage difference of drag and lift coefficients between two subsequent adaptations, which are denoted with ϵ_{C_d} and ϵ_{C_L} respectively. In order to improve the efficiency of the process, these thresholds are lower for coarse meshes and higher for fine meshes. Specifically they are set as

$$\epsilon_{C_d} = [4 \times 10^{-3}, 5 \times 10^{-5}] \quad \epsilon_{C_L} = [2 \times 10^{-2}, 1 \times 10^{-3}]$$

The refinement level of each mesh is based on a number called complexity C (see Section III.A), which is proportional to the number of vertices. Theoretically, the relation between the number of vertices N and C is:

$$N \approx 1.46 C,$$

but, in practice we generally observe for three-dimensional cases $N \approx 2 C$.

The nine complexities considered are:

$$C = \{10\,000, \quad 20\,000, \quad 40\,000, \quad 80\,000, \quad 160\,000 \quad 320\,000 \quad 640\,000 \quad 1\,280\,000 \quad 2\,560\,000\}.$$

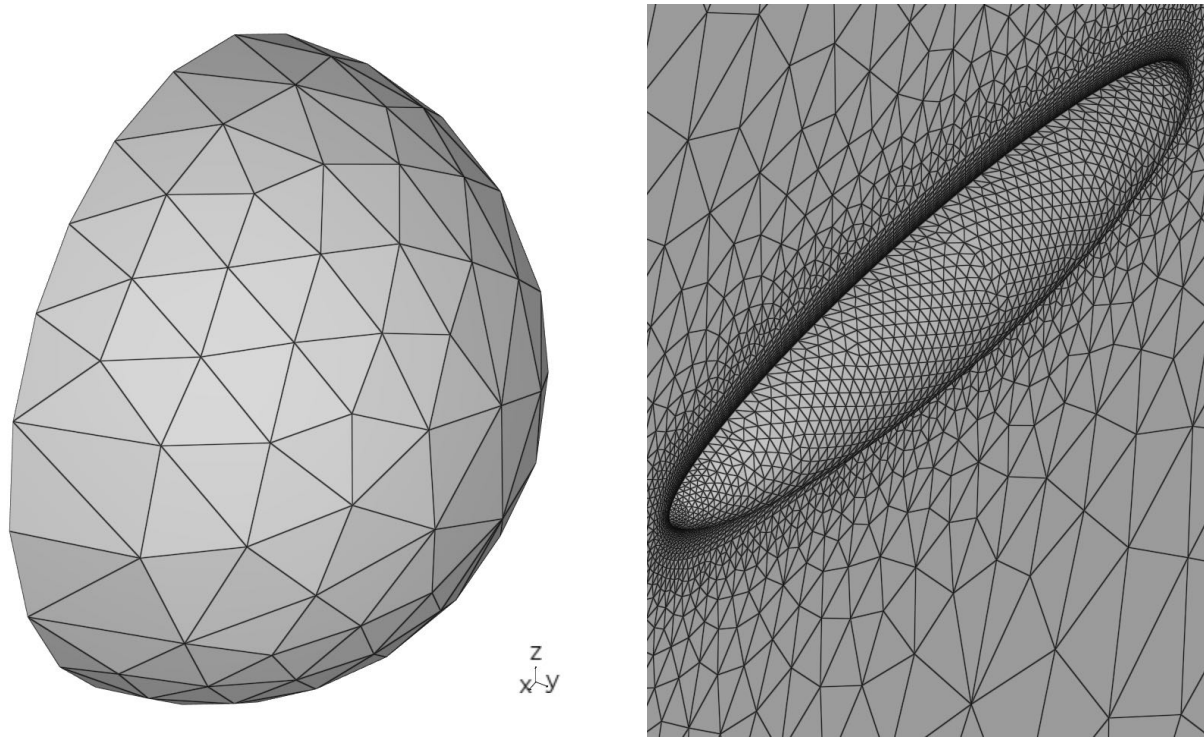


Fig. 12 3D subsonic flow past the prolate spheroid. Initial coarse mesh only composed of 11 527 vertices, 47 202 tetrahedra and 16 158 triangles on the boundaries. Exterior (left) and interior (right) view.

The resulting first and final adapted meshes are composed of 20K and 5.2M vertices.

The adapted mesh and solution obtained for the SA-neg model with 2.6M vertices are shown in Fig. 13

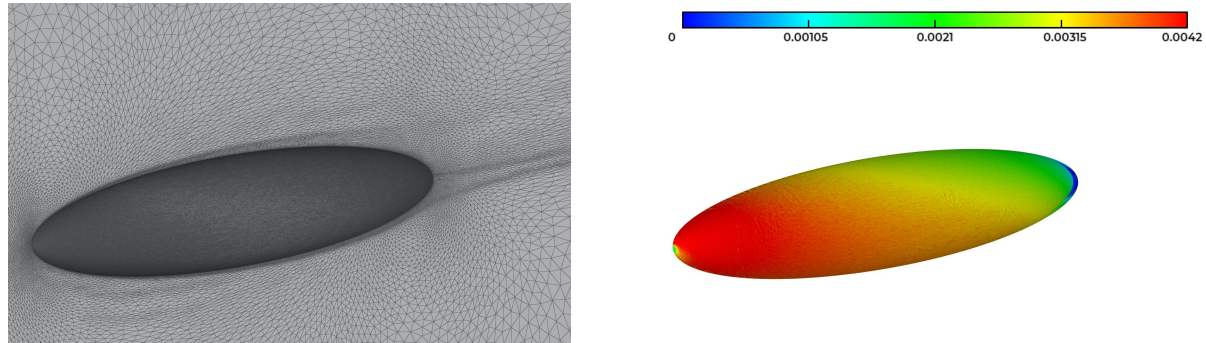


Fig. 13 3D subsonic flow past the prolate spheroid. SA-neg turbulence model. GO anisotropic mesh adaptation based on drag coefficient. Adapted mesh composed of 2 609 771 vertices, 14 104 561 tetrahedra and 1 008 332 triangles on the boundaries. Adapted mesh (left) and skin friction coefficient (right).

We can see that anisotropic mesh adaptation automatically refines the regions of the separated boundary layer and the wake behind the spheroid. One important remark that can be drawn is that even the SA-neg model produces a tiny region of transition at the leading edge. This is clearly a deficiency of the model and this restricted region of transition is purely fictitious.

The mesh-convergence curves for the aerodynamic coefficients are plotted in Fig. 14. The reader can see that mesh-convergence for viscous drag is already achieved on 1M-vertices adapted mesh. Conversely pressure drag results fully converged only on 5.2M-vertices adapted mesh.

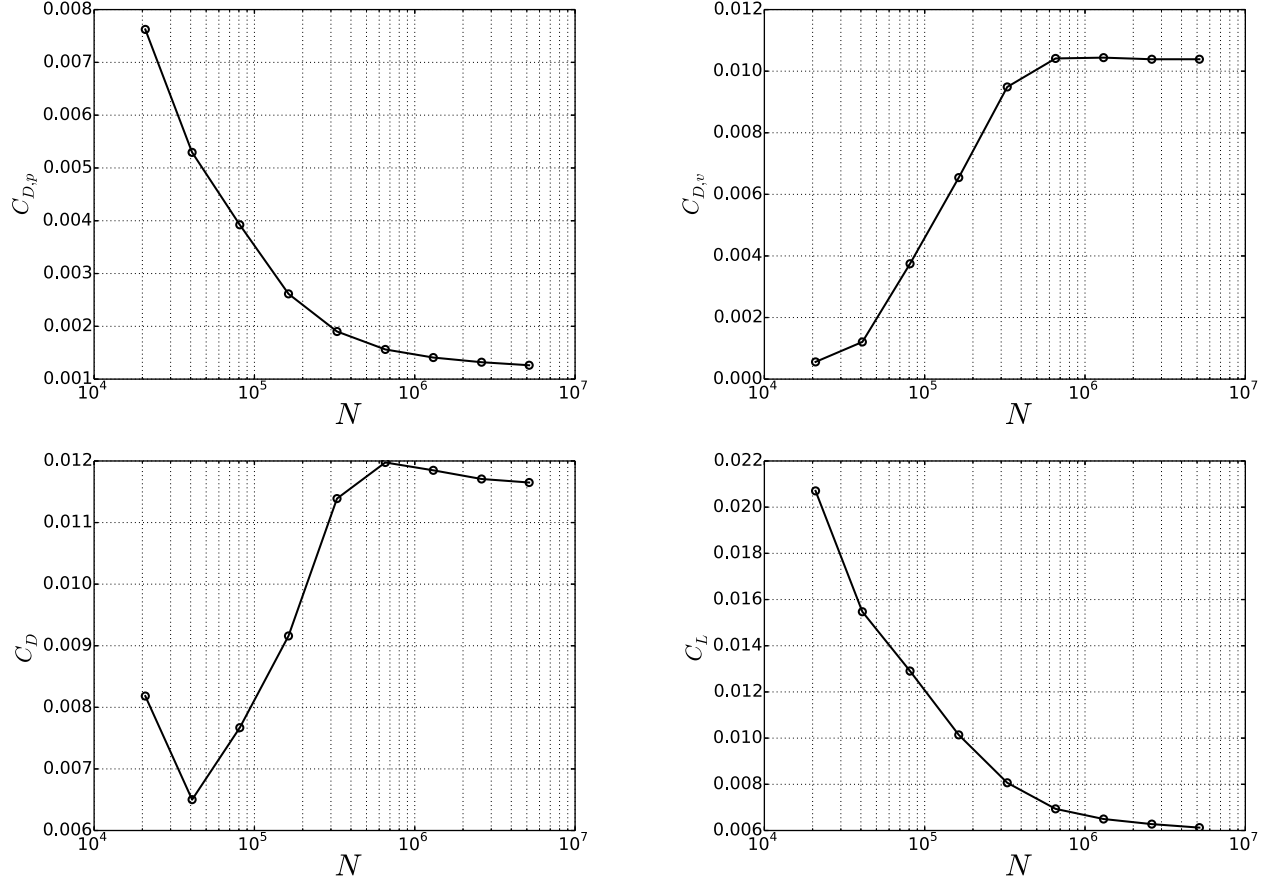


Fig. 14 3D subsonic flow past the prolate spheroid. SA-neg turbulence model. GO anisotropic mesh adaptation based on drag coefficient. Mesh-convergence of pressure drag $C_{D,p}$, viscous drag $C_{D,v}$, drag C_D and lift C_L coefficients. N is the number of vertices.

An analogous mesh-adaptation process using BCM SA-neg model was performed. As previously done, the process is initialized with the SA-neg adapted results obtained on $C = 20\,000$. The adapted mesh and solution obtained for this transitional calculation with 2.6M vertices are shown in Fig. [15]. One first remark that can be drawn is that transition region is already visible in the mesh. In fact, the reader can note that, unlike the fully turbulent SA adapted simulation, the one using the BCM model exhibits an additional layer of refinement near the annular region of transition. In fact, the skin friction field denotes a strong transition which seems occurring almost uniformly along the azimuth direction. To give a more quantitative measure, the skin friction coefficient C_f for both SA model and BCM one, on the cut plane $y = 0$ is plotted in Fig. [16] on the left. Transition is well visible occurring close to $x \approx 0.06$.

Convergence plots for aerodynamic coefficients are plotted in Fig. [17]. Looking at these plots, the reader could quickly say that BCM model experiences the same trend of convergence as SA model and do not require any additionally DoFs w.r.t. a fully turbulent simulation. In order to verify or not this assumption, it worth tracking the convergence of the transition point. In Fig. [16] on the right, the evolution of the transition point through the different meshes produced during the adaptation process is depicted. We can see that even for the complexity $C = 2560\,K$, which corresponds to an adapted mesh composed of 5.2M-vertices, the transition point is still moving downstream. This means that more points are still required to converge the transition region.

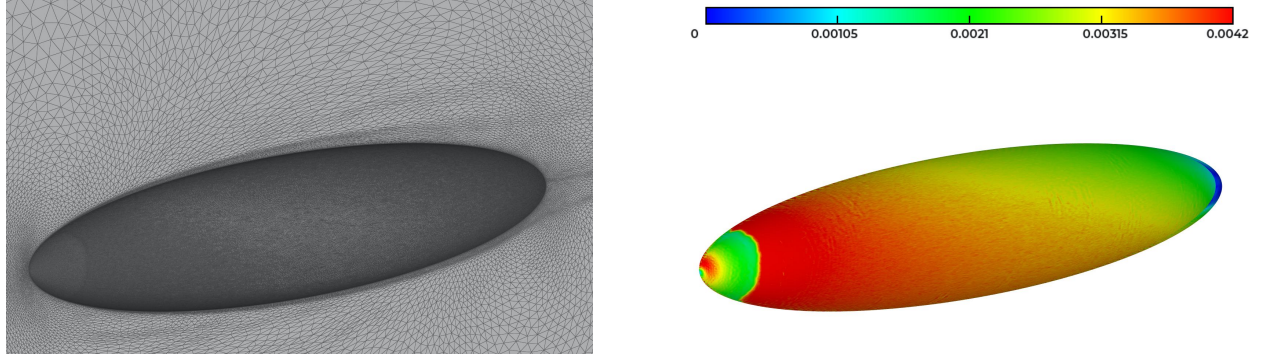


Fig. 15 3D subsonic flow past the prolate spheroid. BCM SA-neg-noft2 turbulence model. GO anisotropic mesh adaptation based on drag coefficient. Adapted mesh composed of 2 598 907 vertices, 14 063 923 tetrahedra and 926 379 triangles on the spheroid surface. Adapted mesh (left) and skin friction coefficient field (right).

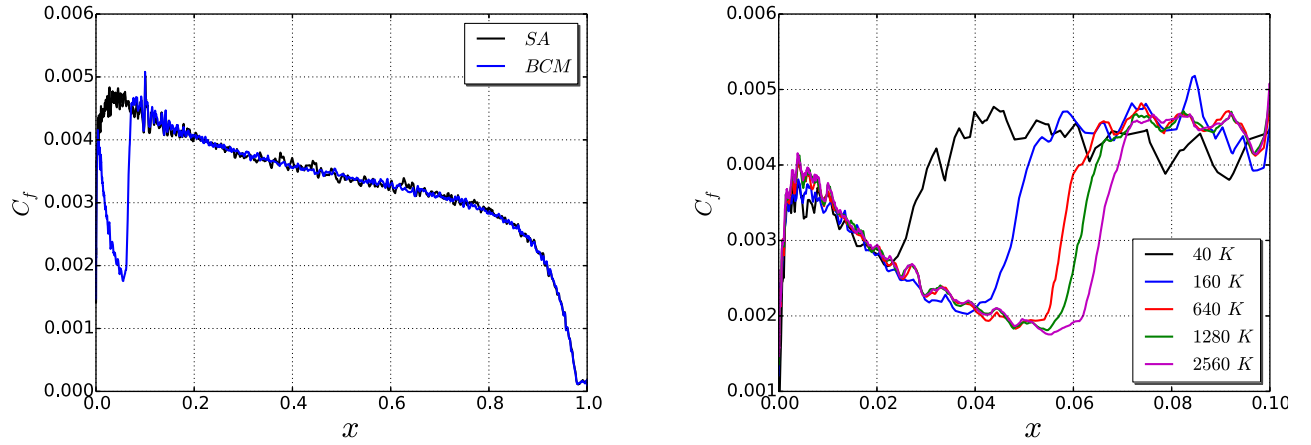


Fig. 16 3D subsonic flow past the prolate spheroid. GO anisotropic mesh adaptation based on drag coefficient. Cut-plane $y = 0$, with x denoting the streamwise coordinate. Skin friction coefficient on final adapted meshes (left) and transition point convergence for different complexities C (right).

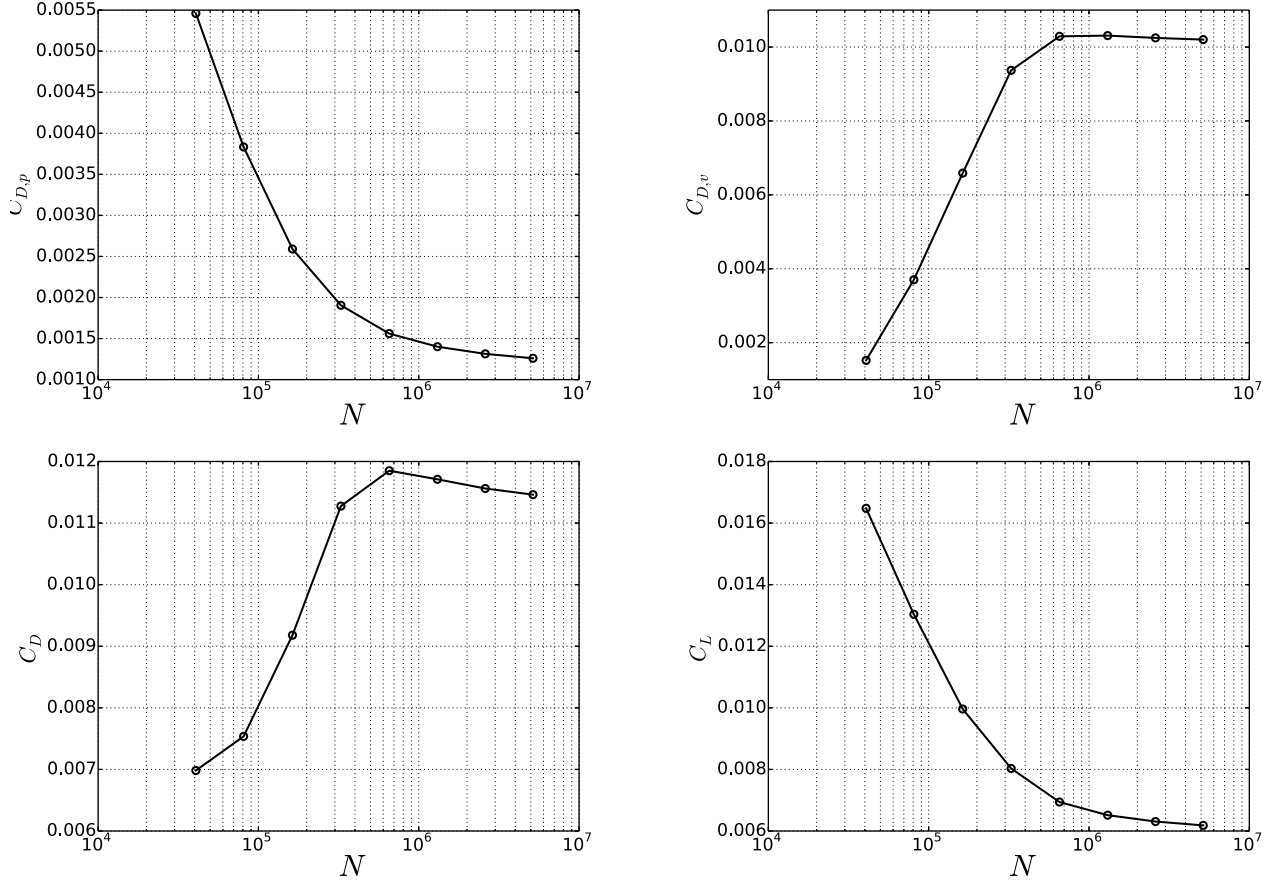


Fig. 17 3D subsonic flow past the prolate spheroid. BCM-noft2-SA-neg turbulence model. GO anisotropic mesh adaptation based on drag coefficient. Mesh-convergence of pressure drag $C_{D,p}$, viscous drag $C_{D,v}$, drag C_D and lift C_L coefficients. N is the number of vertices.

V. Conclusions and Perspectives

The assessment of anisotropic mesh-adaptation for transitional flows with the SA-beg-noft2-BCM model has been presented. The model is proven to be robust for mesh-adaptation processes using both multi-scale error estimate and goal-oriented error estimate. Three pertinent test cases are considered: 2D Zero-Pressure gradient Flat Plate; the 2D subsonic flow past the NLF-0416 airfoil and the 3D subsonic flow past a prolate spheroid.

Transition point mesh-convergence is shown to be achievable with a reasonable number of DoFs on both two-dimensional test cases. Transition point is automatically detected by the error estimate and meshed accordingly. Specifically, transitional adapted calculations seem requiring twice the number of points w.r.t. their corresponding fully turbulent ones. This happens because the transition region is meshed with quasi-isotropic elements, while allowing high levels of anisotropy in the upstream and downstream regions.

Similar considerations can be drawn for three-dimensional calculations. For the subsonic flow past the prolate spheroid, the aerodynamic coefficients show similar mesh-convergence on both fully turbulent and transitional adapted calculations. However, as observed on the previous two-dimensional simulations, higher level of refinement are required to converge the transition point. Specifically, while fully turbulent solution can be considered mesh-converged on a 5 millions vertices-adapted mesh, its corresponding transitional simulation shows a transition point which is still not fully converged with the same mesh size. Even though more refined meshes are required, the asymptotic behavior of the transition point during the mesh adaption process indicate a possible mesh-convergence achievable with 10 millions/20 millions vertices.

These results are very promising in the context of the development of a robust mesh adaptation process for transitional flows. Studies aiming at more complex flows are currently underway and will be reported in subsequent papers.

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